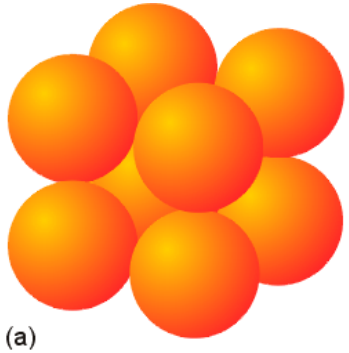
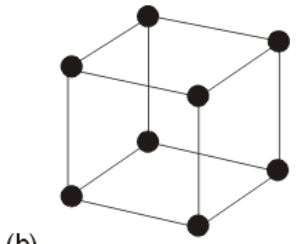
ME 282

Simple Cubic Structure (SC)

- Rare due to poor packing (only Po has this structure)
- The atoms occupy the corners of a cube.
- Close-packed directions are cube edges.
- The coordination number is 6, and the packing efficiency is only 52.4%





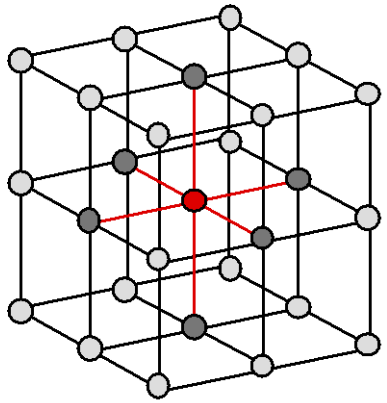


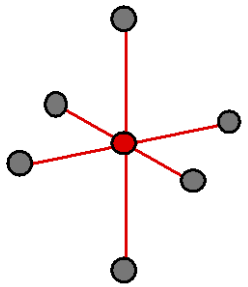
(Courtesy P.M. Anderson)



(a) Primitive cubic metal
1 atom per unit cell

- Coordination # = 6
  (# nearest neighbors)



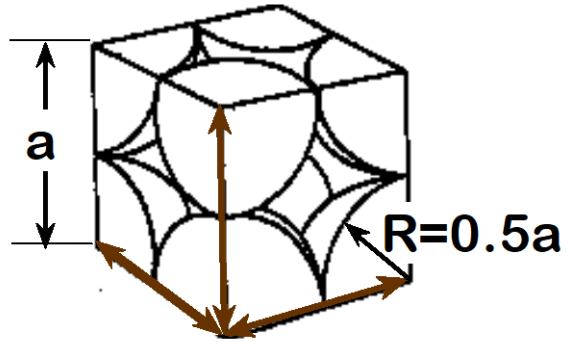


Atomic Packing Factor: SC

$$APF = \frac{\text{Volume of atoms in unit cell}^*}{\text{Volume of unit cell}}$$

*assume hard spheres

- APF for a simple cubic structure = 0.52



close-packed directions

contains 8 x 1/8 =

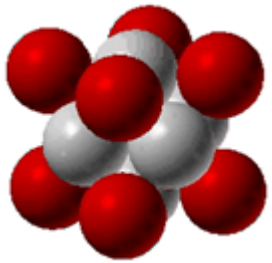
1 atom/unit cell

$$APF = \frac{\frac{\text{atoms}}{\text{unit cell}} \cdot 1 \cdot \frac{4}{3}\pi (0.5a)^3 \cdot \frac{\text{volume}}{\text{atom}}}{a^3 \cdot \frac{\text{volume}}{\text{unit cell}}}$$

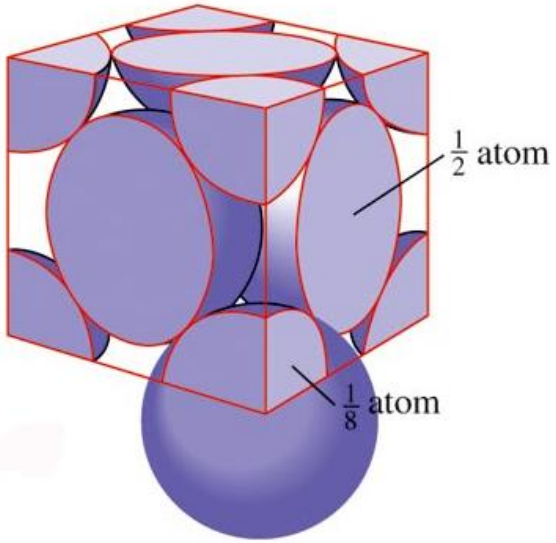
Adapted from Fig. 3.19, *Callister 6e.*

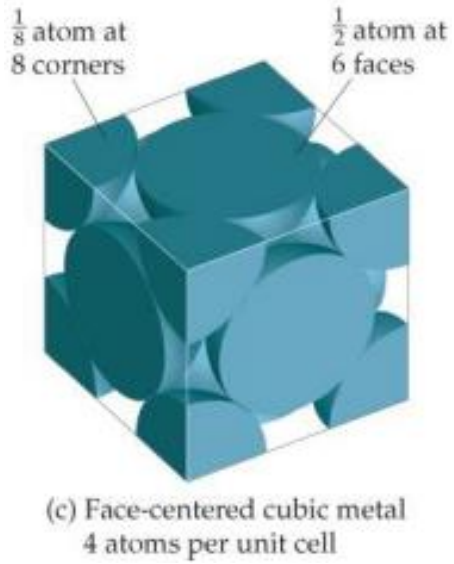
Face-Centered Cubic (FCC)

of Atoms/FCC Unit Cell



- A face-centered cubic unit cell contains a total of 4 atoms: 1 from the corners, and 3 from the faces
- Close packed directions are face diagonals.



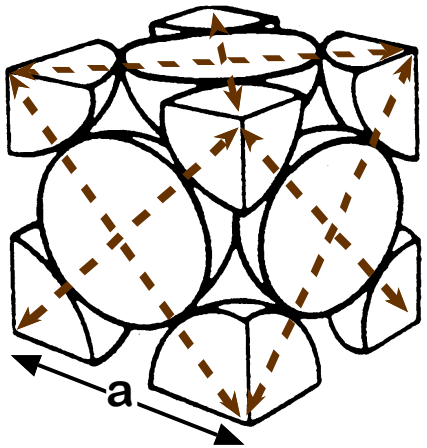


(c) Face-centered cubic metal
4 atoms per unit cell

- Coordination # = 12

Atomic Packing Factor: FCC

- APF for a face-centered cubic structure = 0.74



Close-packed directions:

length = 4R

$= \sqrt{2}\,a$

Unit cell contains:

6 x 1/2 + 8 x 1/8

= 4 atoms/unit cell

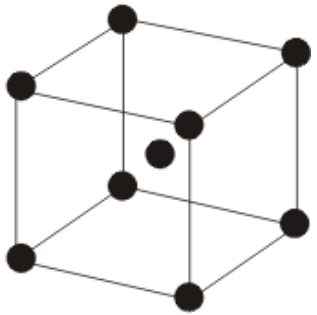
$$APF = \frac{\dfrac{\text{atoms}}{\text{unit cell}} \cdot \dfrac{\text{volume}}{\text{atom}}}{\dfrac{\text{volume}}{\text{unit cell}}} = \frac{4\left(\frac{4}{3}\pi\left(\frac{\sqrt{2}a}{4}\right)^3\right)}{a^3}$$

Adapted from Fig. 3.1(a), *Callister 6e.*

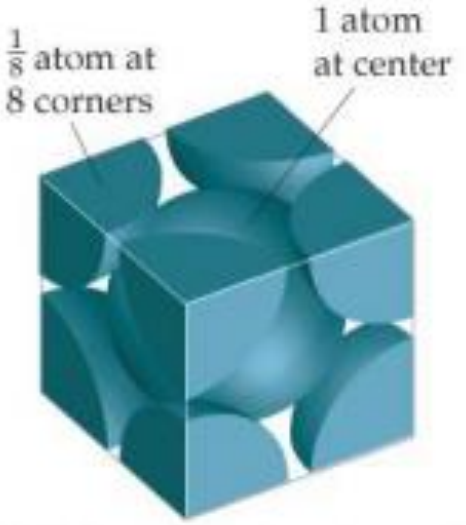
Body-Centered Cubic (BCC)

- *Body-centered cubic* unit cells have an atom in the center of the cube as well as one in each corner
- Close packed directions are cube diagonals
- The packing efficiency is 68%, and the coordination number = 8





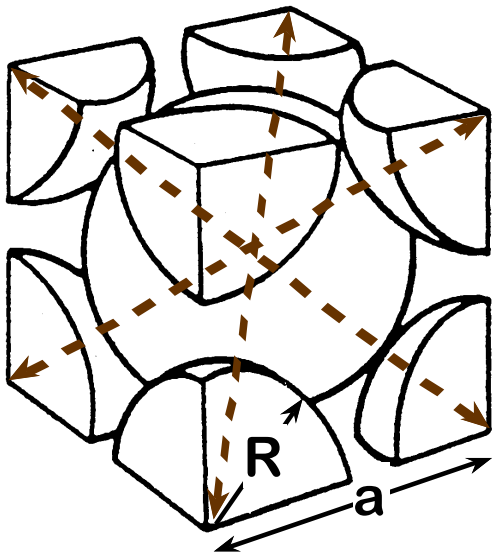
--Note: All atoms are identical; the center atom is shaded differently only for ease of viewing.



(b) Body-centered cubic metal
2 atoms per unit cell

Atomic Packing Factor: BCC

- APF for a body-centered cubic structure = 0.68



Adapted from
Fig. 3.2,
Callister 6e.

Close-packed directions:

length = 4R

$= \sqrt{3}\,a$

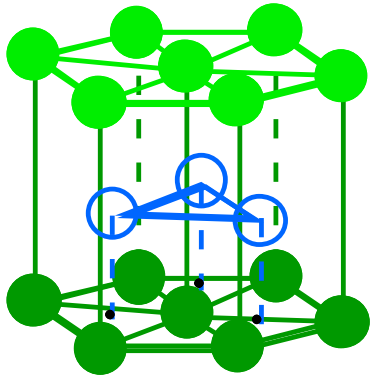
Unit cell contains:

1 + 8 x 1/8

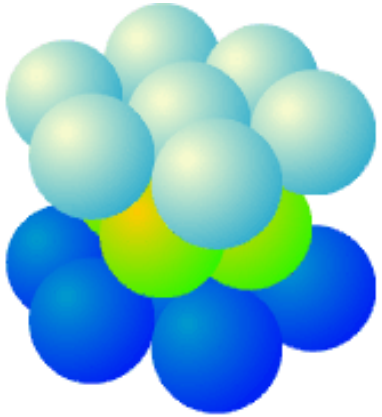
= 2 atoms/unit cell

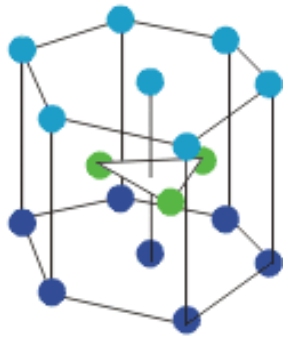
$$APF = \frac{\dfrac{\text{atoms}}{\text{unit cell}} \cdot \dfrac{\text{volume}}{\text{atom}}}{\dfrac{\text{volume}}{\text{unit cell}}} = \frac{2 \cdot \frac{4}{3}\pi\left(\frac{\sqrt{3}a}{4}\right)^3}{a^3}$$

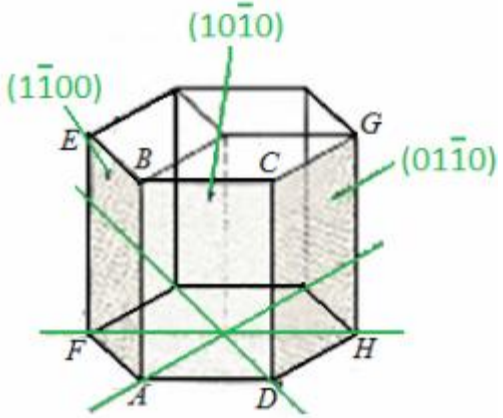
Hexagonal Close-Packed Structure (HCP)



- coordination number = 12
- total 6 atoms per unit cell
- APF = 0.74
- the closest packing possible of spherical atoms
- c/a ratio for ideal HCP structure is 1.633







THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystal Structures - Metallic Crystal Structures

Crystal Structure	a versus R	Atoms per unit cell	APF	Examples
Simple cubic	$a = 2R$	1	0.52	Polonium (Po), α-Mo
BCC	$a = \frac{4R}{\sqrt{3}}$	2	0.68	Fe, W, Mo, Nb, Ta, K, Na, V, Cr
FCC	$a = 2R\sqrt{2}$	4	0.74	Cu, Au, Pt, Ag, Pb, Ni
HCP	$a = 2R$	6	0.74	Ti, Mg, Zn, Be, Co, Zr, Cd

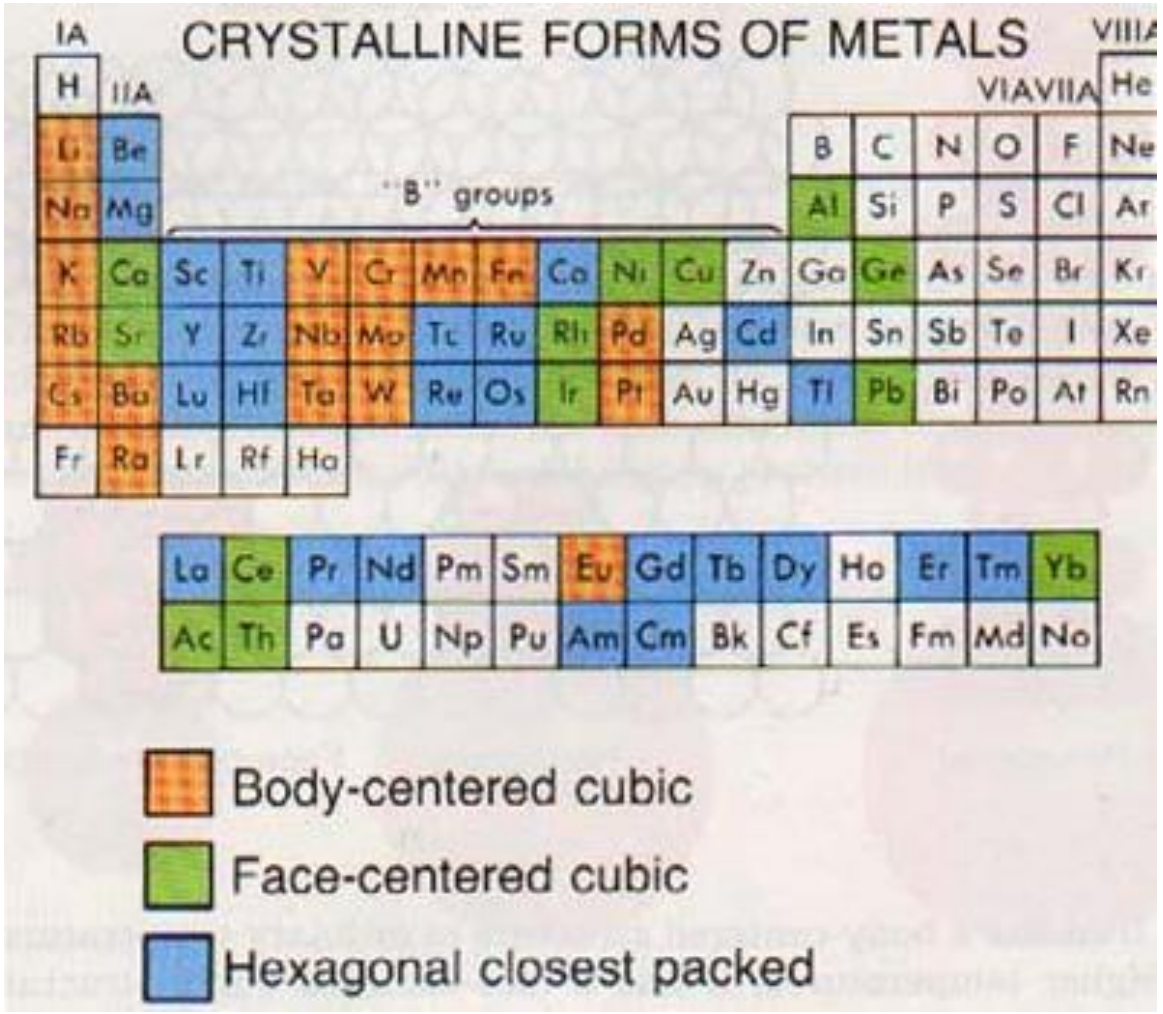
THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystal Structures

Metal	Crystal Structure	Atomic Radius (nm)
Aluminium (Al)	FCC	0.1431
Cadmium (Cd)	HCP	0.1490
Chromium (Cr)	BCC	0.1249
Cobalt (Co)	HCP	0.1253
Copper (Cu)	FCC	0.1278
Gold (Au)	FCC	0.1442
Iron (α) (Fe)	BCC	0.1241
Molybdenum (Mo)	BCC	0.1363

Metal	Crystal Structure	Atomic Radius (nm)
Nickel (Ni)	FCC	0.1246
Platinum (Pt)	FCC	0.1387
Silver (Ag)	FCC	0.1445
Tantalum (Ta)	BCC	0.1430
Titanium (α) (Ti)	HCP	0.1445
Tungsten (W)	BCC	0.1371
Zinc (Zn)	HCP	0.1332
Lead (Pb)	FCC	0.1750

Periodic Table: Close-Packed Structures



Crystal Structures

- Metallic crystal structures tend to be densely packed.
- Reasons for dense packing:
  - Typically, only one element is present, so all atomic radii are the same.
  - Metallic bonding is not directional.
  - Nearest neighbor distances tend to be small in order to lower bond energy.
  - Electron cloud shields cores from each other
- Have the simplest crystal structures (74 elements).

THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystal Structures

Metallic Crystal Structures

❖ *Theoretical Density*

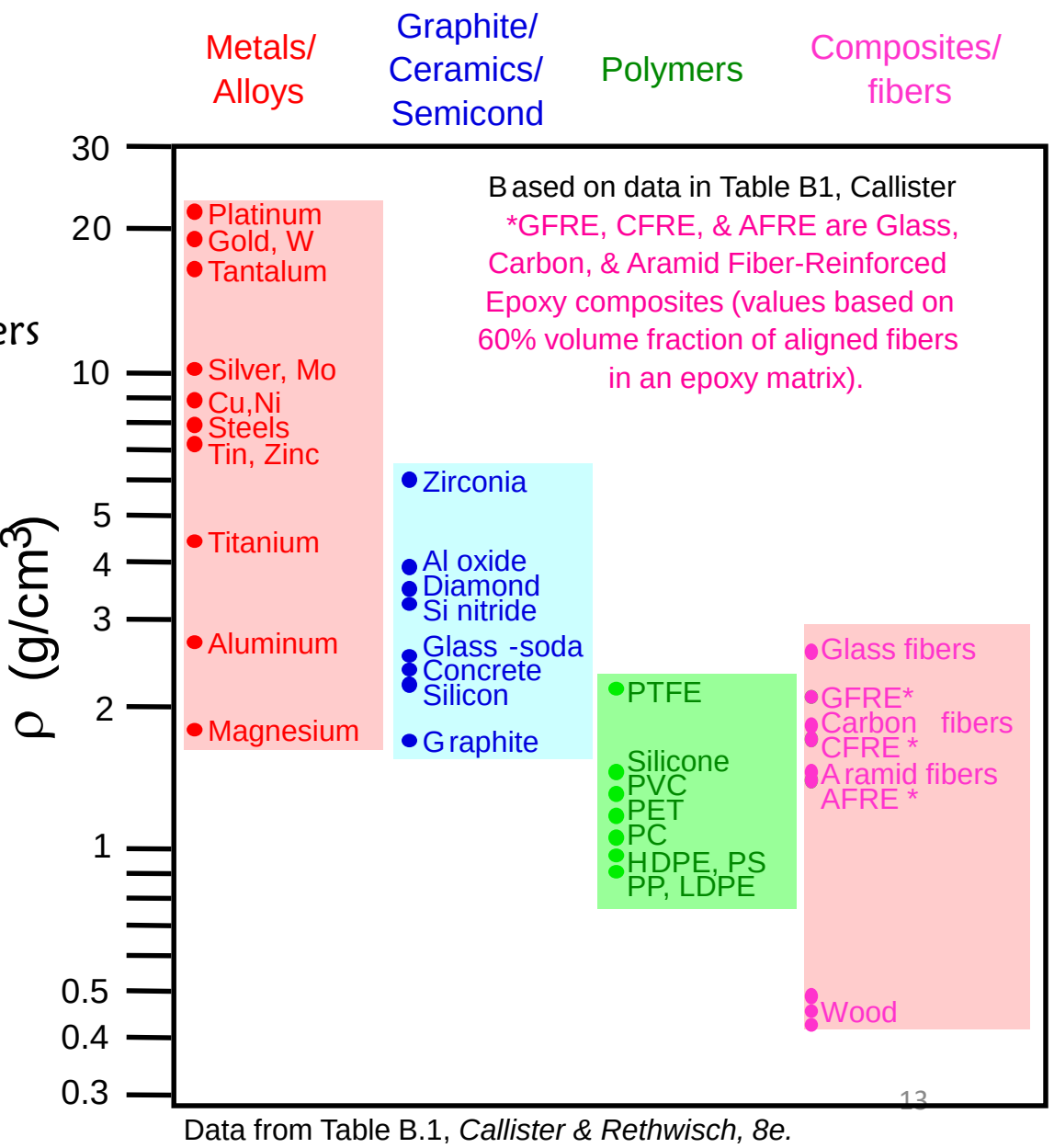
- In general, $\rho_{metals} > \rho_{ceramics} > \rho_{polymers}$

Why?

Metals have
- Close-packing (metallic bonding)
- Often large atomic masses

Ceramics have
- Less dense packing
- Often lighter elements

Polymers have
- low packing density (often amorphous)
- lighter elements



Data from Table B.1, *Callister & Rethwisch, 8e.*

THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystallographic Points, Directions and Planes

- Convention established in which 3 numbers/indices used to designate point coordinates, directions and planes
- Basis for determining index values is
  1. A unit cell
  2. A right-handed coordinate system consisting of (x, y, z) axes with the origin situated at one of the corners and the axes coinciding with the unit cell edges
- Any point within the unit cell can be located
- This is located using the three point coordinate indices: *q*, *r*, and *s*.
- These indices are fractional multiples of the unit cell edge lengths; a, b and c

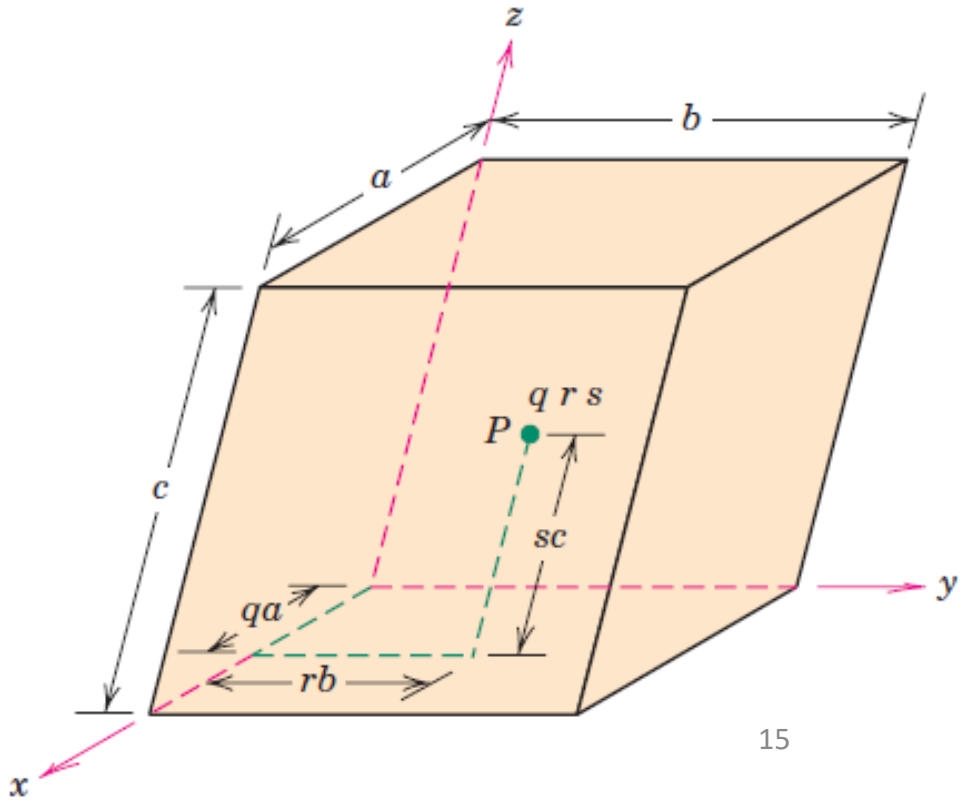
THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystallographic Points, Directions and Planes

Point Coordinates

- *q*, *r* and *s* are the coordinates of point *P* in the unit cell
- *q* is a fraction of edge length *a* i.e. *qa*
- *r* is a fraction of edge length *b* i.e. *rb*
- *s* is a fraction of edge length *c* i.e. *sc*

- Note that by convention, the coordinates of point *P* are written as *q r s* with values less than or equal to unity



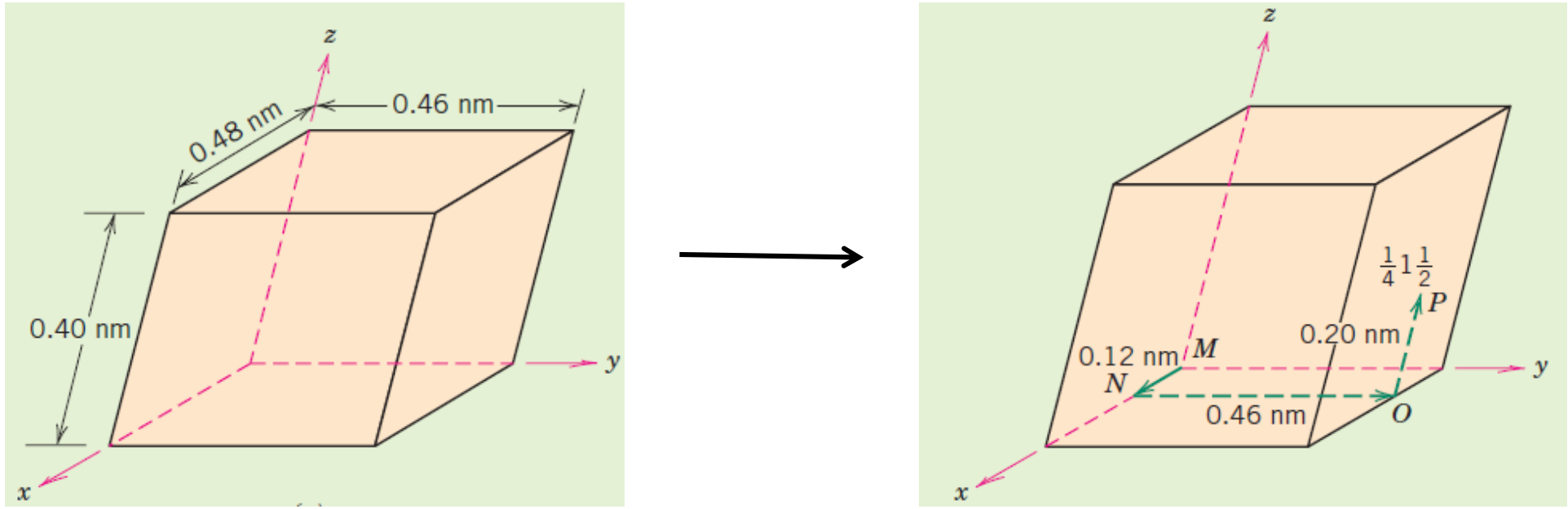
THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystallographic Points, Directions and Planes

Point Coordinates

- For the unit cell shown below, locate the point having coordinates

$$\frac{1}{4}\ 1\ \frac{1}{2}$$

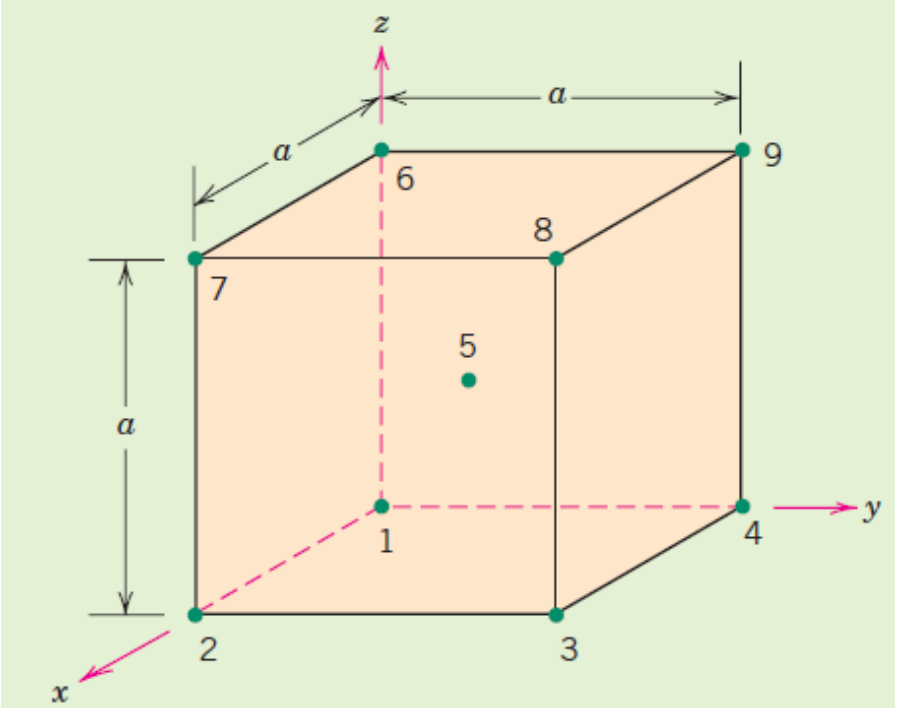


THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystallographic Points, Directions and Planes

Point Coordinates

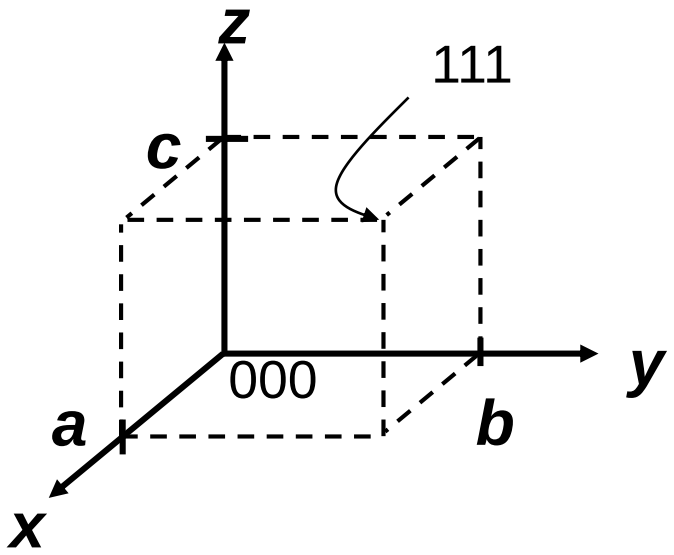
- Specify the point coordinates of all atomic positions in a BCC unit cell in tabular form.

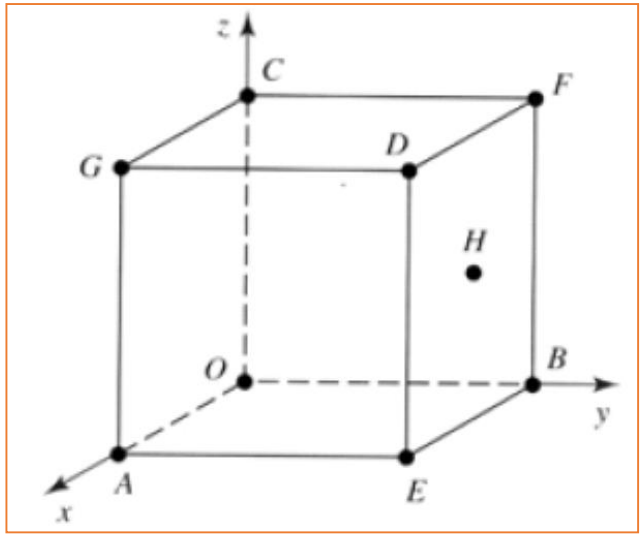


The table must contain the following

Point Number	Fractional Lengths			Point Coordinates
	x axis	*y axis*	*z axis*	
1	0	0	0	0 0 0
2	1	0	0	1 0 0
3	1	1	0	1 1 0
4	0	1	0	0 1 0
5	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}\frac{1}{2}\frac{1}{2}$
6	0	0	1	0 0 1
7	1	0	1	1 0 1
8	1	1	1	1 1 1
9	0	1	1	0 1 1

Locations in Lattices: Point Coordinates





Position	Coordinate
O	0,0,0 (Origin)
A	1,0,0
B	0,1,0
C	0,0,1
D	1,1,1
E	1,1,0
F	0,1,1
G	1,0,1
H	½,1,½

Crystallographic Points, Directions and Planes

Crystallographic Directions

- Directions are lines between two points, or a vector
- The following steps are used to determine the 3 directional indices:

1. A right-handed *x*-*y*-*z* coordinate system is first constructed. As a matter of convenience, its origin may be located at a unit cell corner.
2. The coordinates of two points that lie on the direction vector (referenced to the coordinate system) are determined—for example, for the vector tail, point 1: x_1, y_1, and z_1; whereas for the vector head, point 2: x_2, y_2, and z_2.
3. Tail point coordinates are subtracted from head point components — that is, $x_2 - x_1$, $y_2 - y_1$, and $z_2 - z_1$.
4. These coordinate differences are then normalized in terms of (i.e., divided by) their respective *a*, *b*, and *c* lattice parameters—that is,

Crystallographic Points, Directions and Planes

Crystallographic Directions

4. These coordinate differences are then normalized in terms of (i.e., divided by) their respective *a*, *b*, and *c* lattice parameters – that is,

$$\frac{x_2 - x_1}{a} \quad \frac{y_2 - y_1}{b} \quad \frac{z_2 - z_1}{c}$$

   which yields a set of three numbers

5. If necessary, these three numbers are multiplied or divided by a common factor to reduce them to the smallest integer values.
6. The three resulting indices, not separated by commas, are enclosed in square brackets, thus: [*uvw*]. The *u*, *v*, and *w* integers correspond to the normalized coordinate differences referenced to the *x*, *y*, and *z* axes, respectively.

- In summary, the *u*, *v*, and *w* indices may be determined using the following equations:

$$u = n\left(\frac{x_2 - x_1}{a}\right),\ v = n\left(\frac{y_2 - y_1}{b}\right),\ w = n\left(\frac{z_2 - z_1}{c}\right)$$

THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystallographic Points, Directions and Planes

Crystallographic Directions

Note:

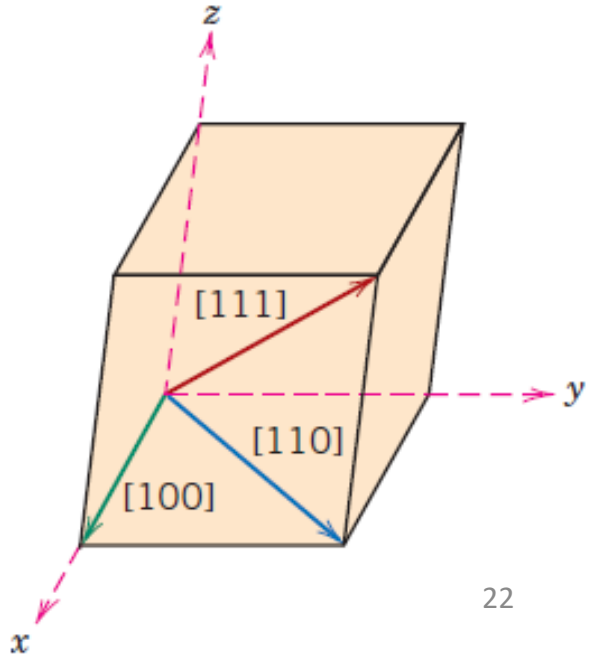
- Negative indices are also possible, which are represented by a bar over the appropriate index
  - $[1\ \bar{1}\ 1]$ has a negative component in the –y direction
- Changing the signs of all the indices produces an anti-parallel direction
  - $[1\ \bar{1}\ 1]$ is directly opposite to $[\bar{1}\ 1\ \bar{1}]$

- Some crystal structures have several nonparallel directions being equivalent i.e. The spacing of atoms along each direction is the same
  - E.g. $[1\ 0\ 0], [\bar{1}\ 0\ 0], [0\ 1\ 0], [0\ \bar{1}\ 0], [0\ 0\ 1]\ and\ [0\ 0\ \bar{1}]$ are all equivalent in cubic crystals

Crystallographic Points, Directions and Planes

Crystallographic Directions

- Equivalent directions are grouped into a ***family***, E.g., a family is written as $\langle 100 \rangle$
- In cubic crystals, directions having the same indices without regard to **order** or **sign** are equivalent
  - E.g. ($[123]$ and $[\bar{2}1\bar{3}]$) are equivalent.



The [100], [110] and [111] directions within a unit cell

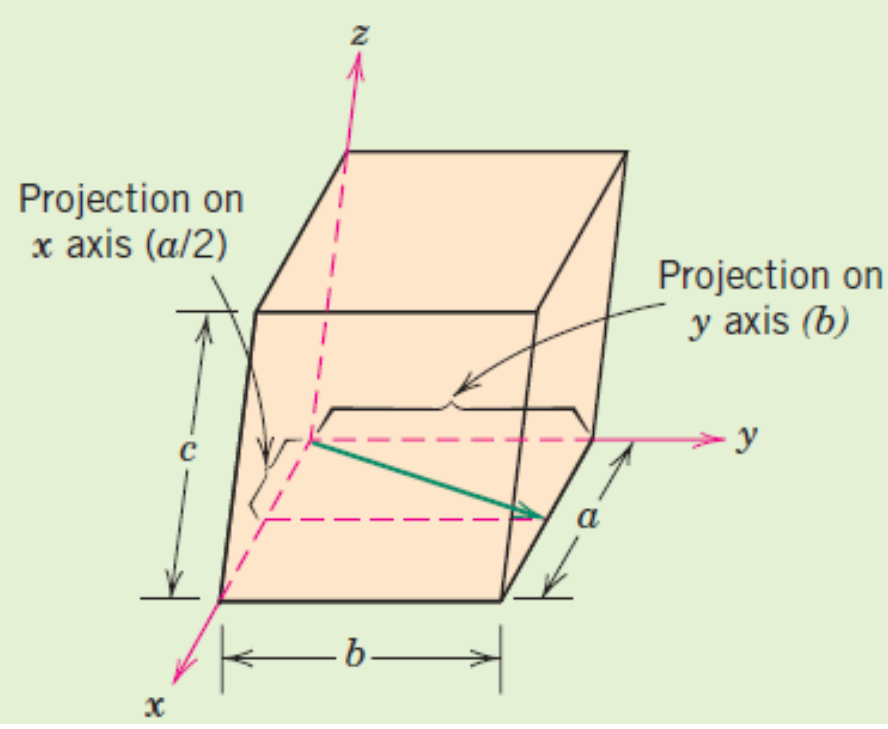
THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystallographic Points, Directions and Planes

Crystallographic Directions

Example:

- Determine the indices of the direction shown below



Summary			
	x	y	z
Projections	**a/2**	**b**	**0c**
Projections in terms of (a, b, c)	**½**	**1**	**0**
Reduction	**1**	**2**	**0**
Enclosure	**[1 2 0]**		

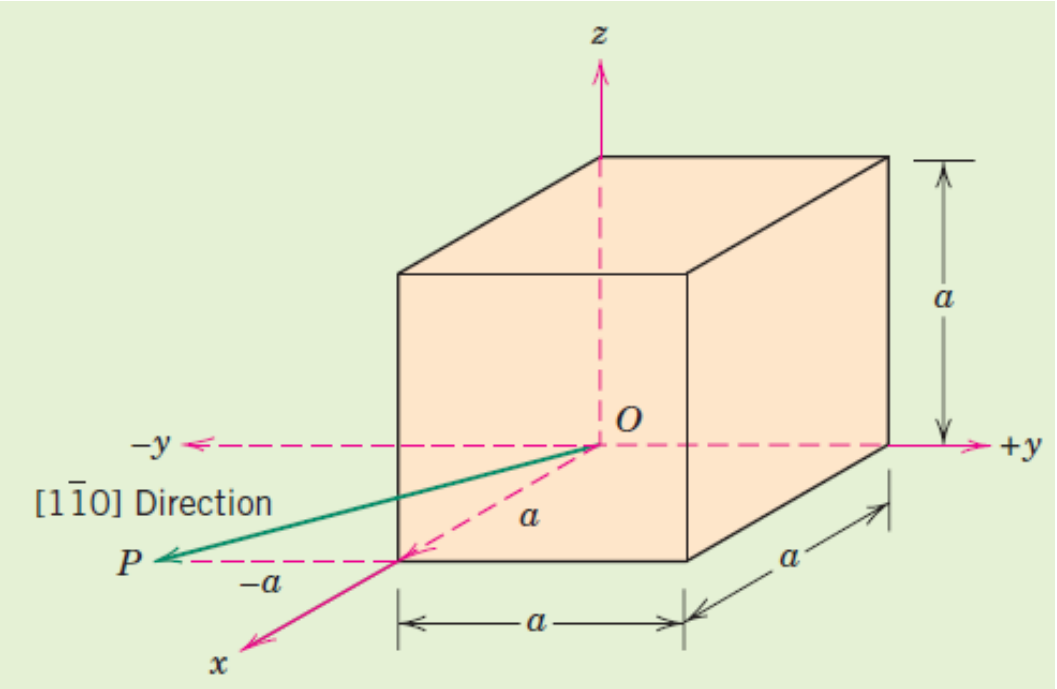
THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystallographic Points, Directions and Planes

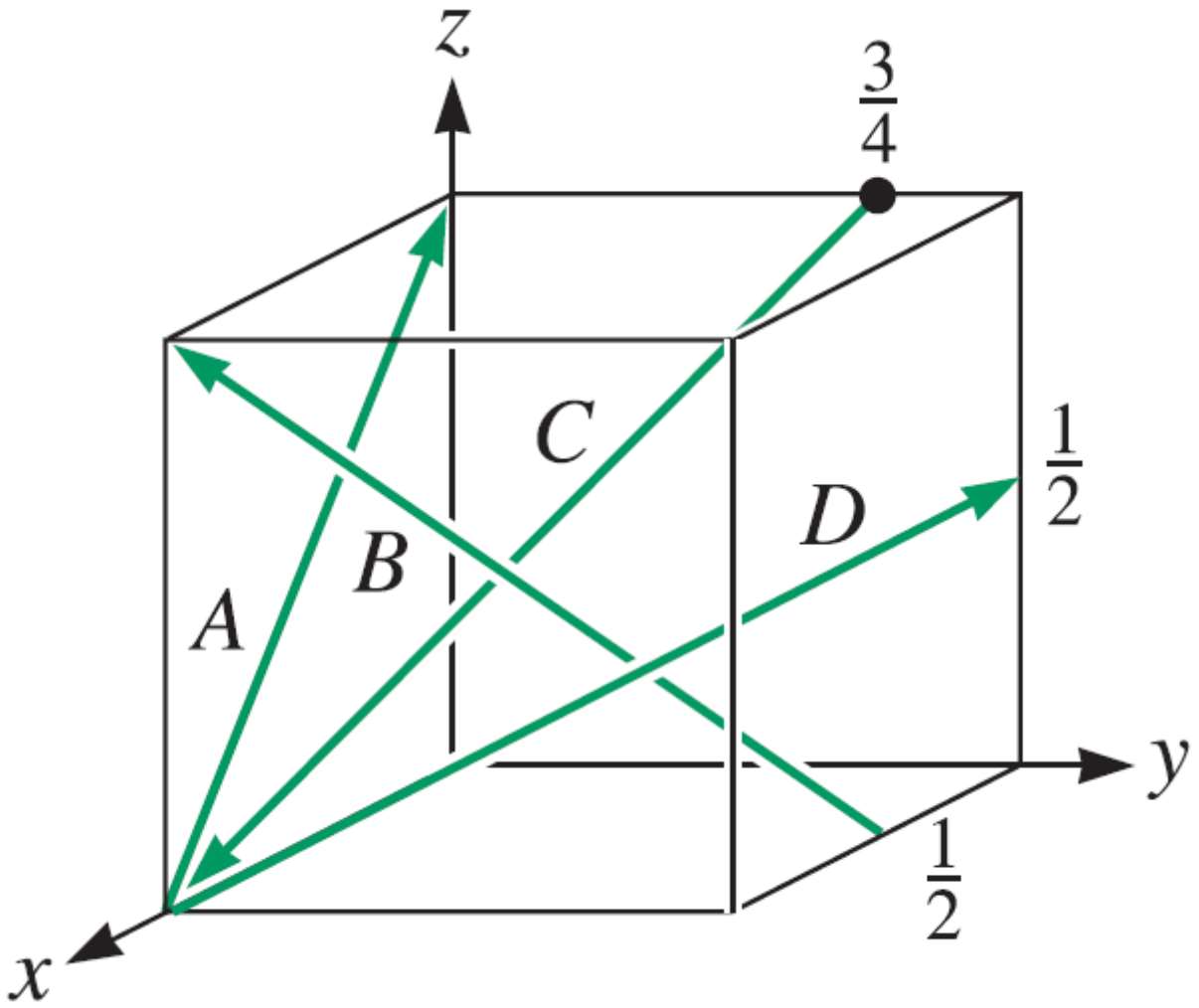
Crystallographic Directions

Example:

- Draw the $[1\ \bar{1}\ 0]$ direction within a cubic unit cell



Determine the indices for the directions in the cubic unit cell



THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystallographic Directions

❖ Hexagonal Crystals

- A four-axis coordinate system otherwise called Miller-Bravais system is employed
- Directional indices are denoted by 4 indices, as ***[u v t w]***
- By convention, the first 3 indices pertain to the projections along the a_1, a_2, a_3 axes in the basal plane
- Conversion from a 3 index system to a 4 index system is achieved using the following formulae

THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystallographic Directions

❖ Hexagonal Crystals

- $[\,u'\,v'\,w'] \rightarrow [\,u\,v\,t\,w]$

$$u = \frac{1}{3}(2u' - v') \qquad v = \frac{1}{3}(2v' - u')$$

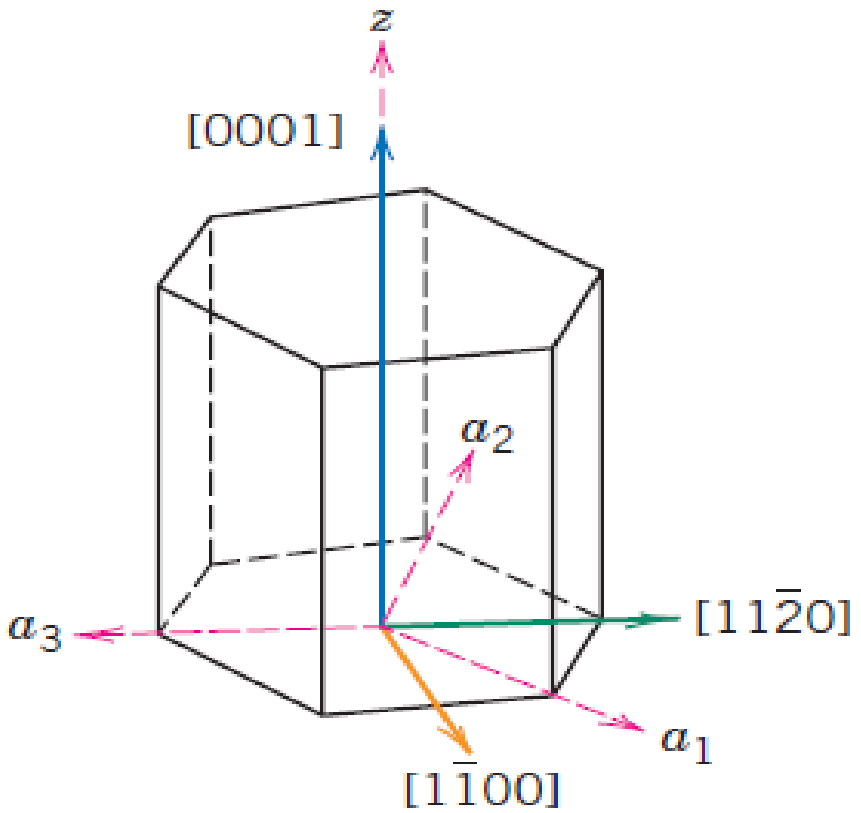
$$t = -(v + u) \qquad w = w'$$

- Primed indices are associated with 3 index scheme
- Unprimed indices are associated with the Miller Bravais
- Four-index system [0 1 0] direction becomes $[\bar{1}\,2\,\bar{1}\,0]$

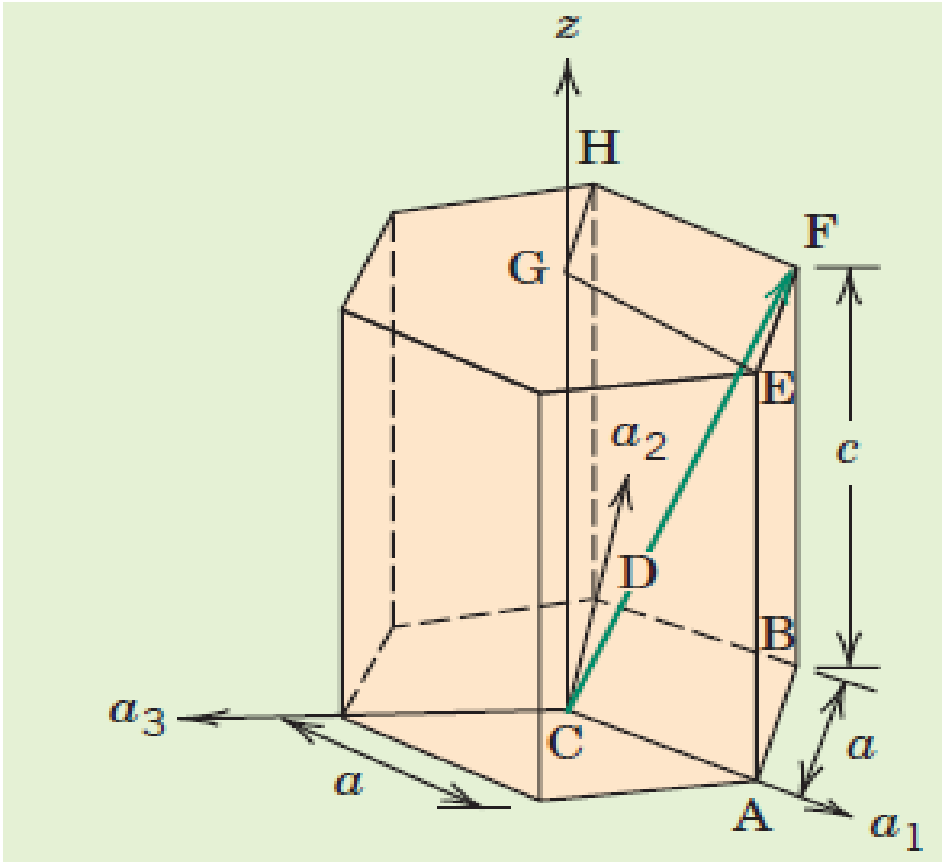
THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystallographic Directions

❖ Hexagonal Crystals



(a)



(b)

THE STRUCTURE OF CRYSTALLINE SOLIDS

Linear Density – For Directions

- Linear density (LD) is defined as the number of atoms per unit length whose centers lie on the direction vector for a specific crystallographic direction, i.e.

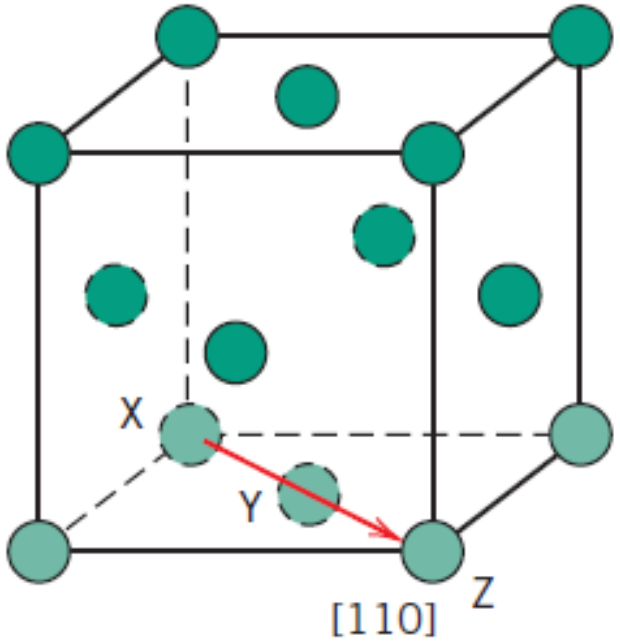
$$LD = \frac{number\ of\ atoms\ centered\ on\ direction\ vector}{length\ of\ direction\ vector}$$

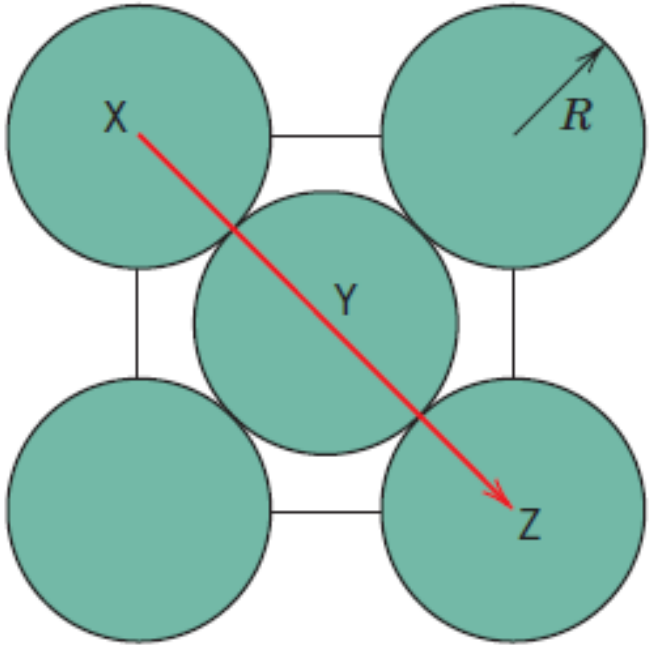
- Units (e.g. nm^{-1}, m^{-1})

THE STRUCTURE OF CRYSTALLINE SOLIDS

Linear Density

- Find the linear density for [1 1 0] within an FCC unit cell





$$LD_{110} = \frac{2 \text{ atoms}}{4R} = \frac{1}{2R}$$